

Apache OpenOffice

Version 4.1

Calc Guide

AOO Documentation Team

Chapter 2

Entering, Editing, and Formatting Data

Copyright

This document is Copyright © 2025 by The Apache Software Foundation. You may distribute it and/or modify it under the terms of the Creative Commons Attribution License, version 3.0 or later (<https://creativecommons.org/licenses/by/3.0/>).

Apache, Apache OpenOffice, and OpenOffice.org are trademarks of the Apache Software Foundation. Used with permission.

Acknowledgments

This document is updated from a previous version by the OOoAuthors team. The AOO Documentation Team appreciates the many contributions of Michael Scarborough and Michele Petrovsky.

Publication date and software version

Published August 25, 2025. Based on Apache OpenOffice 4.1.

Note for Mac users

Some keystrokes and menu items differ on a Mac from those used in Windows and Linux. The table below gives some common substitutions for the instructions in this chapter. For a more detailed list, see the application Help.

Windows/Linux	Mac equivalent	Effect
Tools > Options menu selection	OpenOffice > Preferences	Access setup options
<i>Right-click</i>	<i>Control+click</i>	Open context menu
<i>Ctrl</i> (Control)	⌘ (Command)	Used with other keys
<i>F5</i>	<i>Shift+⌘+F5</i>	Open the Navigator
<i>F11</i>	⌘+T	Open Styles & Formatting window

Contents

Chapter 2

Entering, Editing, and Formatting Data.....	1
Introduction.....	4
Entering Data Using the Keyboard.....	4
Entering Numbers.....	4
Entering Text.....	4
Entering Numbers as Text.....	4
Entering Dates and Times.....	5
Entering Special Characters.....	5
Deactivating Automatic Changes.....	6
Speeding up Data Entry.....	7
Using the Fill Tool on Cells.....	7
Using Selection Lists.....	10
Sharing Content between Sheets.....	10
Validating Cell Contents.....	11
Editing Data.....	13
Removing Data from a Cell.....	13
Replacing All the Data in a Cell.....	14
Changing Part of the Data in a Cell.....	14
Formatting Data.....	14
Formatting Multiple Lines of Text.....	15
Shrinking Text to Fit the Cell.....	16
Formatting Numbers.....	16
Formatting the Font.....	17
Choosing Font Effects.....	18
Setting Cell Alignment and Orientation.....	18
Formatting the Cell Borders.....	20
Formatting the Cell Background.....	20
Autoformatting Cells and Sheets.....	21
Defining a New AutoFormat.....	21
Formatting Spreadsheets Using Themes.....	22
Using Conditional Formatting.....	22
Hiding and Showing Data.....	23
Outline Group Controls.....	24
Filtering Which Cells are Visible.....	25
Sorting Records.....	26
Finding and Replacing in Calc.....	28
Using the Find Toolbar.....	28
Using the Find & Replace Dialog.....	28
Finding and Replacing Formulas or Values.....	29
Finding and Replacing Text.....	30
Finding and Replacing Cell Styles.....	30
Using Wildcards (Regular Expressions).....	30

Introduction

You can enter data into Calc in several ways:

- with the keyboard
- with dragging and dropping with the mouse
- with the Fill tool
- with selection lists.

Calc also allows users to enter information simultaneously into multiple sheets of the same document. Data can also be read from a database or text-based files like CSV (comma-separated values).

Entering Data Using the Keyboard

Most data entry in Calc can be accomplished using the keyboard.

Entering Numbers

Click in the cell and type in the number using the number keys on either the main keyboard or the numeric keypad.

To enter a negative number, either type a minus (-) sign in front of it, or enclose it in parentheses (brackets), like this: (1234).

By default, numbers are right-aligned, and negative numbers have a leading minus symbol.

Entering Text

Click in the cell and type the text. Text is left-aligned by default.

Entering Numbers as Text

If a number is entered in the format *01481*, Calc will drop the leading 0. (One exception is discussed in the Tip below.) To preserve a leading zero, as some contexts like telephone numbers might require, type an apostrophe before the number, for example, '01481. When you do that, the data in question is treated as text and displayed exactly as entered. Typically, calculations carried out by formulas will treat the entry as a zero or ignore it.

Tip

Numbers can have leading zeros and still be regarded as numbers (as opposed to text) if the cell is formatted appropriately. Right-click on the cell and choose **Format Cells > Numbers**. Adjust the Leading zeros setting to add leading zeros to numbers.

Note

When a plain apostrophe is used to allow a leading 0 to be displayed, it is not visible in the cell after the Enter key is pressed and it's not treated as part of the cell's contents. For example, it will not be detected by the Find & Replace function. If "smart quotes" are used for apostrophes, the apostrophe remains visible in the cell.

To choose this type of apostrophe, use **Tools > AutoCorrect Options > Localized Options**. Select the **Replace** option for apostrophes to activate this function. But be advised that whatever apostrophe type you settle upon will affect both Calc and Writer.

Caution



When a number is formatted as text, take care that the cell containing the number is not used in a formula. In that scenario, Calc will simply ignore the value.

Entering Dates and Times

Select the cell and type the date or time. You can separate the date elements with a slash (/) or a hyphen (-) or use text such as 10 Oct 03. Calc recognizes a variety of date formats. You can separate time elements with colons, such as 10:43:45.

Entering Special Characters

A "special" character is one not found on a standard English keyboard. For example, © ¼ æ ç ñ ö ø ¢ are all special characters. To insert a special character:

- 1) Place the cursor where you want the character to appear in your document.
- 2) Click **Insert > Special Character** to open the Special Characters dialog (Figure 1).
- 3) Select the characters (from any font or mixture of fonts) you wish to insert in order; then click **OK**. The selected characters are shown in the bottom left of the dialog. As you select each character, it is shown alone at the bottom right, along with the numerical code for that character.

Note

Different fonts include different special characters. If you do not find a particular special character you want, try changing the *Font* selection.

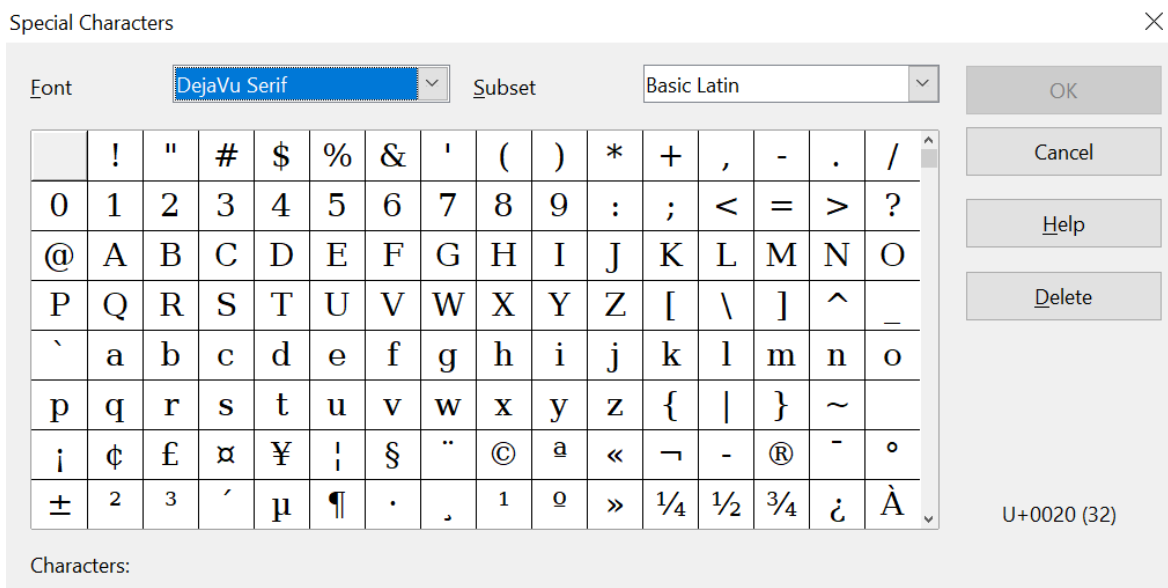


Figure 1: The Special Characters dialog

Inserting Dashes

To enter **en** and **em** dashes, you can use the *Replace dashes* option under **Tools > AutoCorrect Options > Options** tab. Under certain conditions, this option replaces two hyphens with the corresponding dash.

In the following table, the A and B represent text consisting of letters A to Z or digits 0 to 9.

Text that you type:	Result
A - B (A, space, minus, space, B)	A - B (A, space, en-dash, space, B)
A -- B (A, space, minus, minus, space, B)	A - B (A, space, en-dash, space, B)
A--B (A, minus, minus, B)	A—B (A, em-dash, B)
A-B (A, minus, B)	A-B (unchanged)
A -B (A, space, minus, B)	A -B (unchanged)
A --B (A, space, minus, minus, B)	A -B (A, space, en-dash, B)

Deactivating Automatic Changes

Calc automatically applies many changes during data input unless you deactivate those changes. You can also immediately undo any automatic changes with **Ctrl+Z**.

AutoCorrect changes

Automatic correction of typing errors, replacement of straight quotation marks by curly (custom) quotes, and starting cell content with an uppercase (capital) letter are controlled by **Tools > AutoCorrect Options**. Go to the *Options* or *Replace* tabs to deactivate any features you do not want. On the *Replace* tab, you can also delete unwanted word pairs and add new ones as required.

AutoInput

When you are typing in a cell, Calc automatically suggests matching input found in the same column. To turn the AutoInput on and off, set or remove the check mark in front of **Tools > Cell Contents > AutoInput**.

Automatic date conversion

Calc automatically converts certain entries to dates. To ensure that an entry that looks like a date is interpreted as text type an apostrophe at the beginning of the entry.; the apostrophe is not displayed in the cell.

Speeding up Data Entry

Entering data into a spreadsheet can be labor-intensive, but Calc provides several tools for removing some of the drudgery from input.

The most basic of these is to drop and drag the contents of one cell to another with a mouse. Many people also find AutoInput helpful.

Calc includes several other tools for automating input, especially of repetitive material. They include the Fill tool, selection lists, and the ability to input information into multiple sheets of the same document.

Using the Fill Tool on Cells

At its simplest, the Fill tool duplicates existing content. Start by selecting the cell to copy, then drag the mouse in any direction (or hold down the Shift key and click in the last cell you want to fill), and then choose **Edit > Fill** and the direction in which you want to copy: Up, Down, Left, or Right.

Caution



Unavailable choices are grayed out, but you can still choose the opposite direction from what you intend. That could cause you to overwrite cells accidentally.

Tip

A shortcut to fill cells is to grab the “handle” in the lower right-hand corner of the cell and drag it in the direction you want to fill. If the cell contains a number, the number will fill in series. If the cell contains text, the same text will fill in the direction you chose. If the cell contains a number and you hold the *Ctrl* key while dragging the handle, the number will not increment within the fill area.

	A	B
1	Original	
2		
3		
4		
5		
6		
7		
8		
9		
10		
11		
12		

	A	B
1	Original	
2	Original	
3	Original	
4	Original	
5	Original	
6	Original	
7	Original	
8	Original	
9	Original	
10	Original	
11		
12		

Figure 2: Using the Fill tool

Using a Fill Series

A more complex use of the Fill tool is to use fill series. The default lists are for the full and abbreviated days of the week and the months of the year, but you can also create your own lists.

To add a fill series to a spreadsheet, select the cells to fill and choose **Edit > Fill > Series**. In the Fill Series dialog, select **AutoFill** as the *Series type* and enter an item from any defined series as the *Start value*. The selected cells then fill in the other items on the list sequentially, repeating from the top when they reach the end of the list.

Fill Series

Direction: ☒ Down, ☐ Right, ☐ Up, ☐ Left

Series type: ☐ Linear, ☐ Growth, ☐ Date, ☒ AutoFill

Time unit: ☒ Day, ☐ Weekday, ☐ Month, ☐ Year

Start value: January

End value:

Increment: 1

Buttons: OK, Cancel, Help

Figure 3: Specifying the start of a fill series (result is in Figure 4)

	A	B
1	January	
2	February	
3	March	
4	April	
5	May	
6	June	
7	July	
8	August	
9	September	
10	October	
11	November	
12	December	
13		

Figure 4: Result of fill series selection shown in Figure 3

You can also use **Edit > Fill > Series** to create a one-time fill series for numbers by entering the start and end values and the increment. For example, if you entered start and end values of 1 and 7 with an increment of 2, you would get the sequence of 1, 3, 5, 7.

In such cases, the Fill tool creates only a momentary connection between the cells. Once they are filled, the cells have no further connection with one another.

Defining a Fill Series

To define your own fill series, go to **Tools > Options > OpenOffice Calc > Sort Lists**. This dialog shows the previously defined series in the *Lists* box on the left and the contents of the highlighted list in the *Entries* box.

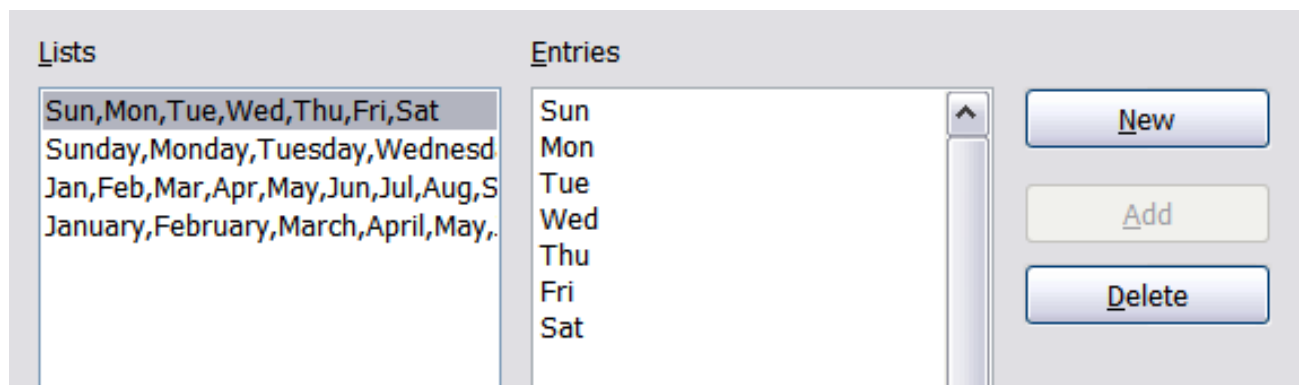


Figure 5: Predefined fill series

Click **New**. The *Entries* box is cleared. Type the series for the new list in the *Entries* box (one entry per line), then click **Add**.

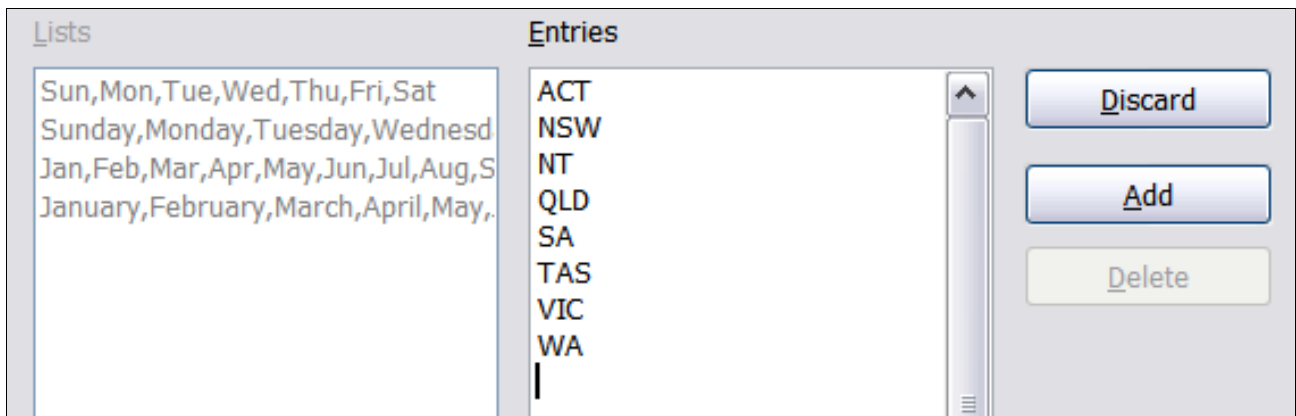
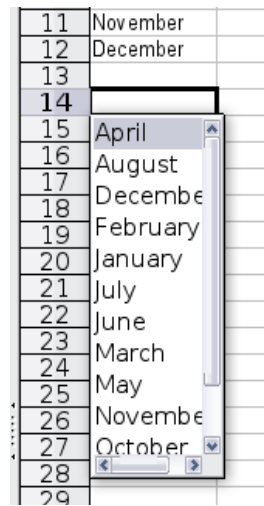


Figure 6: Defining a new fill series

Using Selection Lists

Selection lists are available only for text and are limited to using only text that has already been entered in the same column.

To use a selection list, select a blank cell and press **Ctrl+D**. A drop-down list appears on any cell in the same column with at least one text character or whose format is defined as text. Click on the entry you require.



Sharing Content between Sheets

You might want to enter the same information in the same cell on multiple sheets, for example, to set up standard listings for a group of individuals or organizations. Instead of entering the list on each sheet individually, you can enter it in all the sheets at once. To do this, select all the sheets with **Edit > Sheet > Select** and using **Shift + Click** or **Ctrl + Click**, then enter the information in the current one.

Caution



This technique overwrites any information already in the cells on the other sheets—without any warning. For this reason:

- be careful with this tool
- when you are finished, be sure to deselect all the sheets except the one you want to edit. (**Ctrl+click** on a sheet tab to select or deselect the sheet.)

Validating Cell Contents

When creating spreadsheets for other people to use, you may want to make sure they enter data that is valid or appropriate for the cell. You can also use validation in your own work as a guide to entering data that is either complex or rarely used.

Fill series and selection lists can handle some types of data but are limited to predefined information. To validate new data entered by a user, select a cell and use **Data > Validity** to define the type of contents that can be entered in the cell. For example, a cell might require a date or a whole number, with no alphabetic characters or decimal points, or a cell can be defined so that it cannot be left empty.

Depending on how validation is set up, the tool can also define the range of values that can be entered and provide help messages that:

- explain the content rules you have set up for the cell
- explain what users should do when they enter invalid content.

You can also set the cell to refuse invalid content, accept it with a warning, or—if you are especially well-organized—start a macro when an error is entered.

Validation is most useful for cells containing functions. If cells are set to accept invalid content with a warning, rather than refusing it, you can use **Tools > Detective > Mark Invalid Data** to find the cells with invalid data. The Detective function marks any cells containing invalid data with a circle.

Note that a validity rule is considered part of a cell's format. If you select **Format** or **Delete All** from the Delete Contents window, the rule is removed. To copy a validity rule with the rest of the cell, use **Edit > Paste Special > Paste Formats** or **Paste All**.

Figure 7 shows the choices for a typical validity test. Note the **Allow blank cells** option under the *Allow* list.

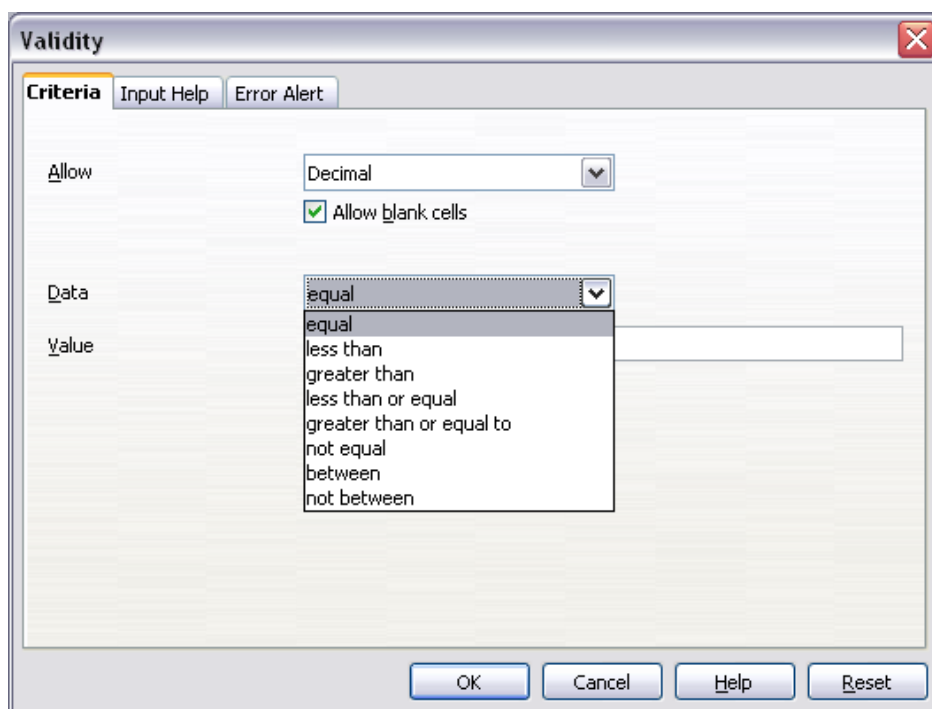


Figure 7: Typical validity test choices

Validity test options vary with the type of data selected from the *Allow* list. For example, Figure 8 shows the choices when a cell must contain a value from a cell range.

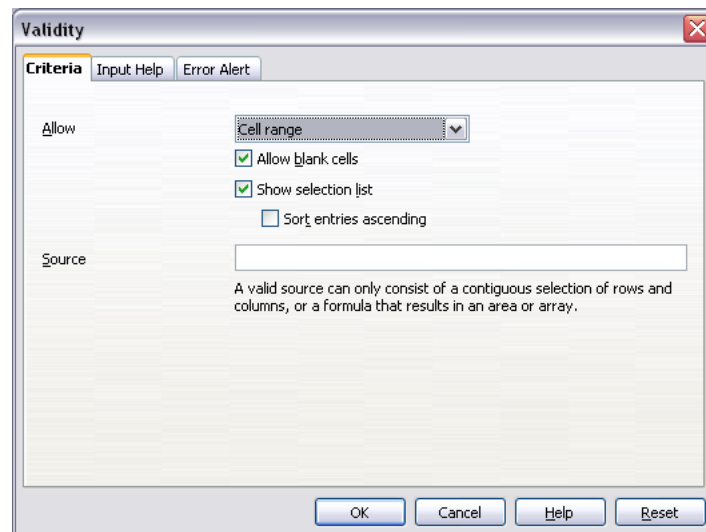


Figure 8: Validity choices for a cell range

To provide input help for a cell, use the Input Help tab of the Validity dialog (Figure 9). To show an error message when an invalid value is entered, use the Error Alert tab (Figure 10). Be sure to write a helpful explanation of what a valid entry should contain. Provide more than just a simple “invalid data—try again” message. To be of real help to users, particularly inexperienced ones, error messages need to provide details not only on what went wrong but also on how to correct it.

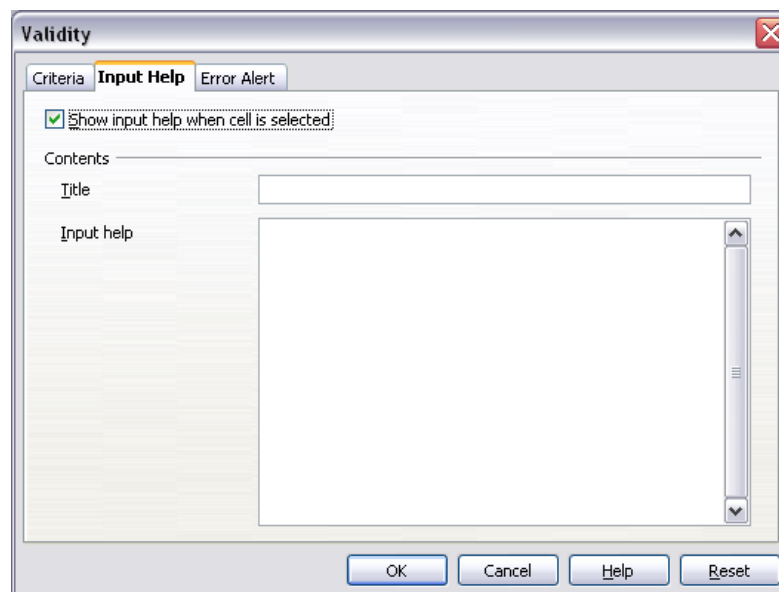


Figure 9: Defining input help for a cell

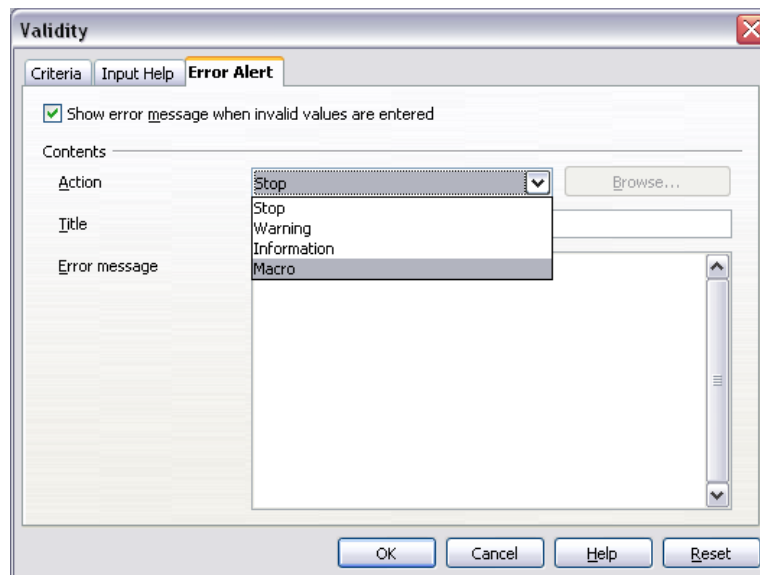


Figure 10: Defining an error message for a cell with invalid data

Editing Data

Editing data is very similar to entering it. The first step is to select the cell containing the data to be edited.

Removing Data from a Cell

Data can be removed (deleted) from a cell in several ways.

Removing data only

Data can be removed from a cell without removing any of the cell's formatting. Click in the cell to select it; then press the *Backspace* key.

Removing data and formatting

Data and its formatting can be removed from a cell simultaneously. Press the *Delete* key (or right-click and choose **Delete Contents**, or use **Edit > Delete Contents**) to open the *Delete Contents* dialog (Figure 11). From this dialog, different aspects of the cell can be deleted. To delete everything in a cell (contents and format), check **Delete all**.

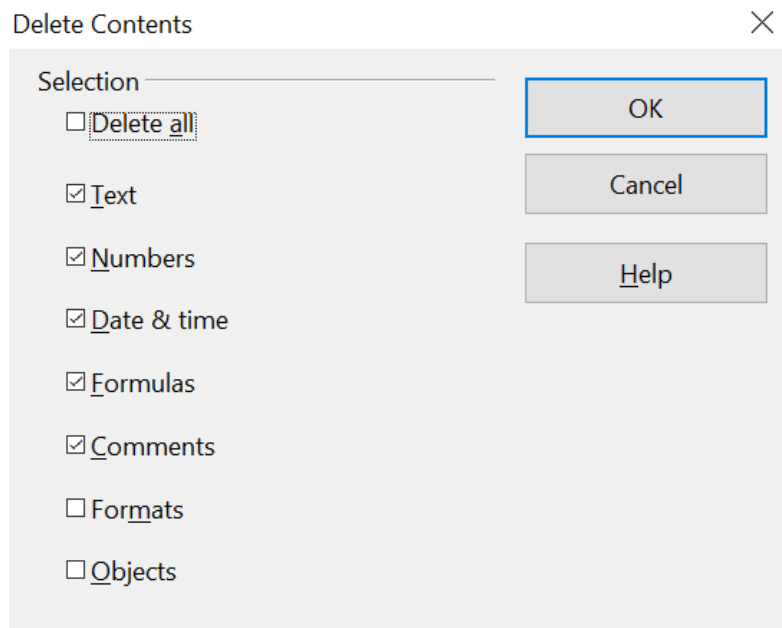


Figure 11: Delete Contents dialog

Replacing All the Data in a Cell

To remove data and insert new items, simply type over the old data. The new data will retain the original formatting.

Changing Part of the Data in a Cell

Sometimes, it's necessary to change the contents of a cell without removing all the contents. As an example, consider the phrase “See Dick run” in a cell that needs to be changed to “See Dick run fast.” It's often useful to do tasks like this without deleting old cell contents first.

The process is similar to the one described above. The difference in this case is that you need to place the cursor inside the cell. You can do this in two ways.

Using the keyboard

After selecting the appropriate cell, press the *F2* key, and the cursor will be placed at the end of the cell. Then, use the keyboard arrow keys to move the cursor through the text in the cell.

Using the mouse

Using the mouse, either double-click on the appropriate cell (to select it and place the cursor in it for editing) or single-click to select the cell, move the mouse pointer up to the input line, and click into it to place the cursor for editing.

Formatting Data

The data in Calc can be formatted in several ways. It can be edited as part of a cell style so that it is automatically applied or applied manually to the cell. Some manual formatting can be applied using icons on the Properties deck of the Sidebar or on toolbars. For more control and extra options, select the appropriate cell or cell range and click the *More Options* icon at the top right of any of the panels of the Sidebar's Properties deck. You can also right-click on the selected cell or cell range and select **Format Cells**. All of the format options are discussed below.

Note

All the settings discussed in this section can also be set as a part of the cell style. See Chapter 4 (Using Styles and Templates) for more information.

Formatting Multiple Lines of Text

Multiple lines of text can be entered into a single cell using automatic wrapping or manual line breaks. Each of these methods is useful and can be applied to different situations.

Using Automatic Wrapping

To set text to wrap at the end of the cell, right-click on the cell and select **Format Cells** (or choose **Format > Cells** from the menu bar, or press **Ctrl+1**). On the **Alignment** tab (Figure 13), under Properties, select **Wrap text automatically**. The results are shown below (Figure 12).

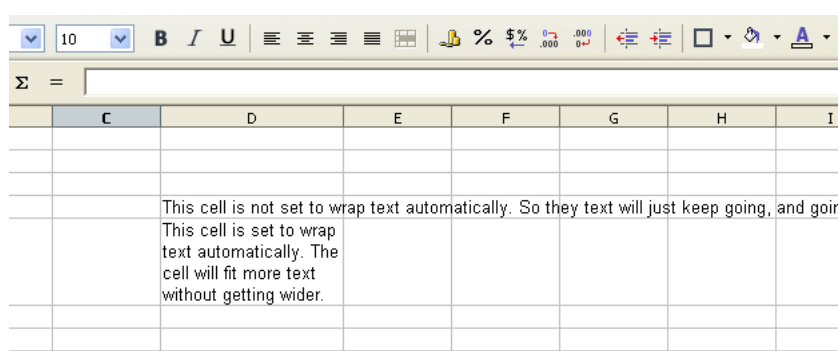


Figure 12: Automatic text wrap

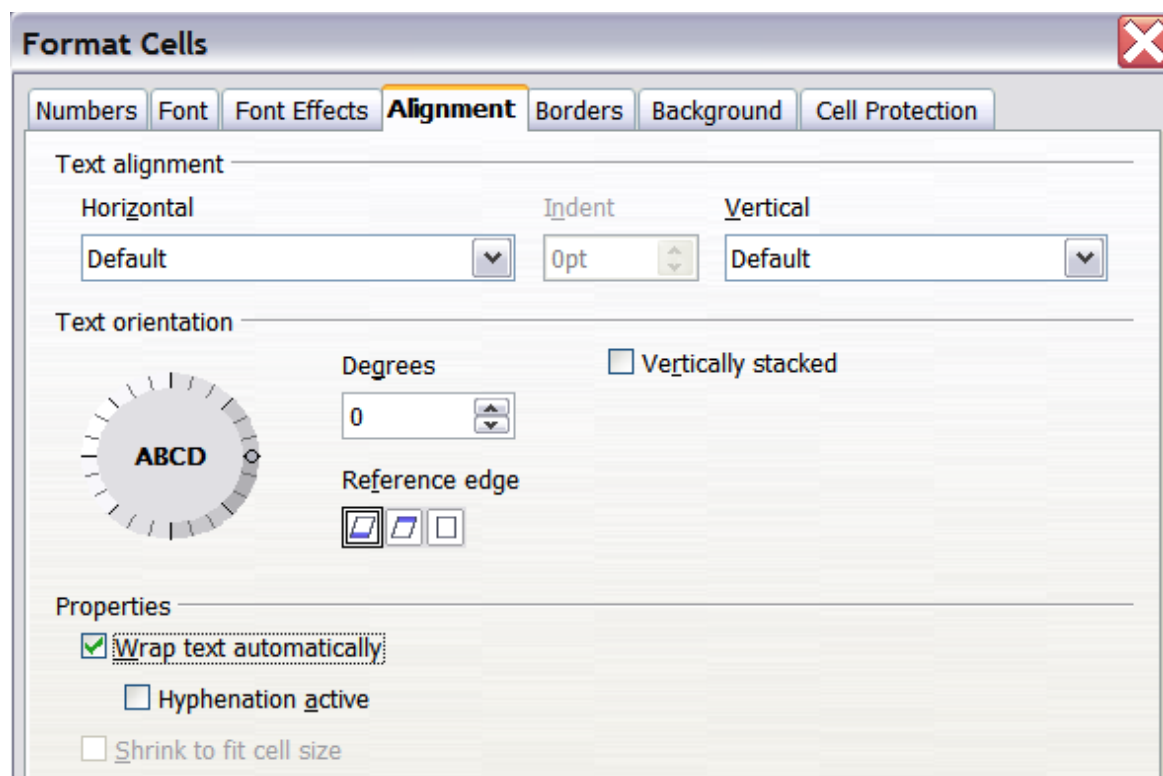


Figure 13: Format Cells > Alignment dialog

Using Manual Line Breaks

To insert a manual line break at the current cursor position while typing in a cell, press **Ctrl+Enter**. Be aware that this method does not work with the cursor in the input line; you must be typing in the cell. When editing text, first double-click the cell, then single-click at the position where you want the line break.

When a manual line break is entered, the cell width does not change. Figure 14 shows the results of using two manual line breaks after the first line of text.

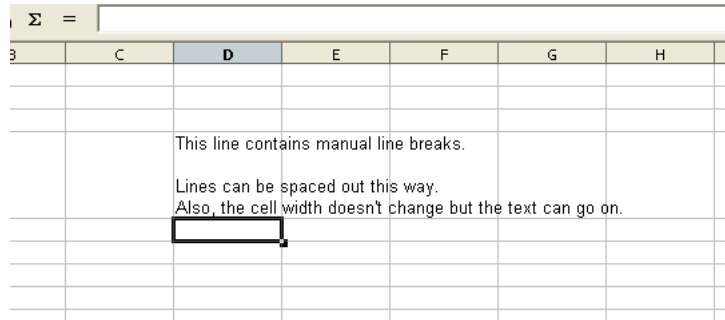


Figure 14: Cell with manual line breaks

Shrinking Text to Fit the Cell

The font size of the data in a cell can automatically adjust to fit in a cell. To do this, select the **Shrink to fit cell size** option in the Format Cells dialog (Figure 13). Figure 15 shows the results.

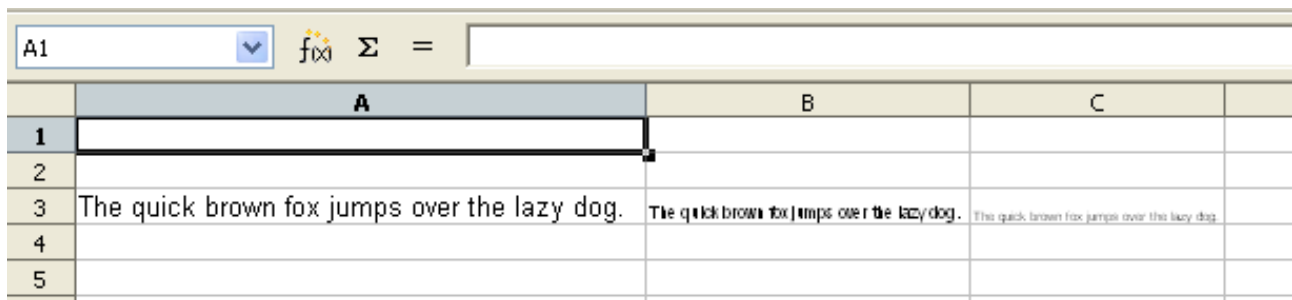


Figure 15: Shrinking font size to fit cells

Formatting Numbers

Several different number formats can be applied to cells using icons on either the Sidebar or the Formatting toolbar. Select the cell, then click the relevant icon. Some icons may not be visible in a default setup; click the down arrow at the end of the Formatting bar and select other icons to display.



Figure 16: Number format icons. Left to right: currency, percentage, date, exponential, standard, add decimal place, delete decimal place.

Use the **Numbers** tab (Figure 17) of the Format Cells dialog for more control, or to select other number formats.

- Apply any of the data types in the Category list to the data.
- Control the number of decimal places and leading zeros.

- Enter a custom format code.

The Language setting controls the local settings for the different formats, such as the date order and the currency marker.

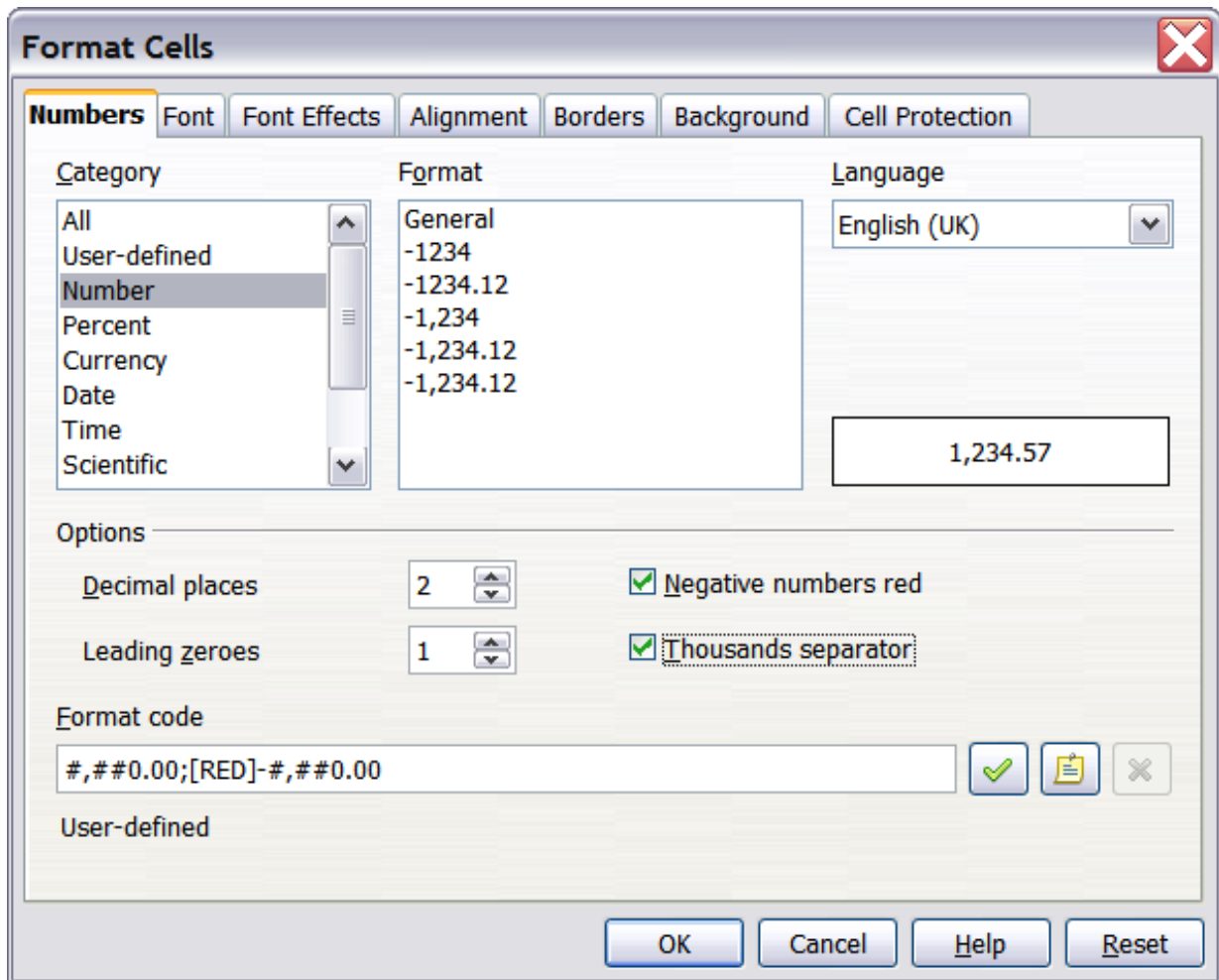


Figure 17: Format Cells > Numbers

Formatting the Font

To quickly choose the font used in a cell:

- select the cell
- click the arrow next to the **Font Name** box on the Formatting toolbar
- choose a font from the list

Tip

To choose whether to show the font names in their font or in plain text, go to **Tools > Options > OpenOffice > View** and select or deselect the Show preview of fonts option in the Font Lists section. For more information, see Chapter 14 (Setting Up and Customizing Calc).

To choose font size, click the arrow next to the Font Size box on the Sidebar's Text panel or the Formatting toolbar. For other formatting, you can use the Bold, Italic, or Underline icons.

To choose a font color, click the arrow next to the **Font Color** icon to display a color palette. Click on the required color.

(To define custom colors, use **Tools > Options > OpenOffice > Colors**. See Chapter 14 for more information.)

To specify the language of the cell (useful because it allows different languages to exist in the same document and be spell-checked correctly), use the *Font* tab of the Format Cells dialog. See Chapter 4 for more information.

Choosing Font Effects

The *Font Effects* tab (Figure 18) of the Format Cells dialog offers more font options.

Overlining and underlining

You can choose from various overlining and underlining options (solid lines, dots, short and long dashes, in various combinations) and the color of the line.

Strikethrough

The strikethrough options include lines, slashes, and Xs.

Relief

The relief options are embossed (raised text), engraved (sunken text), outline, and shadow.

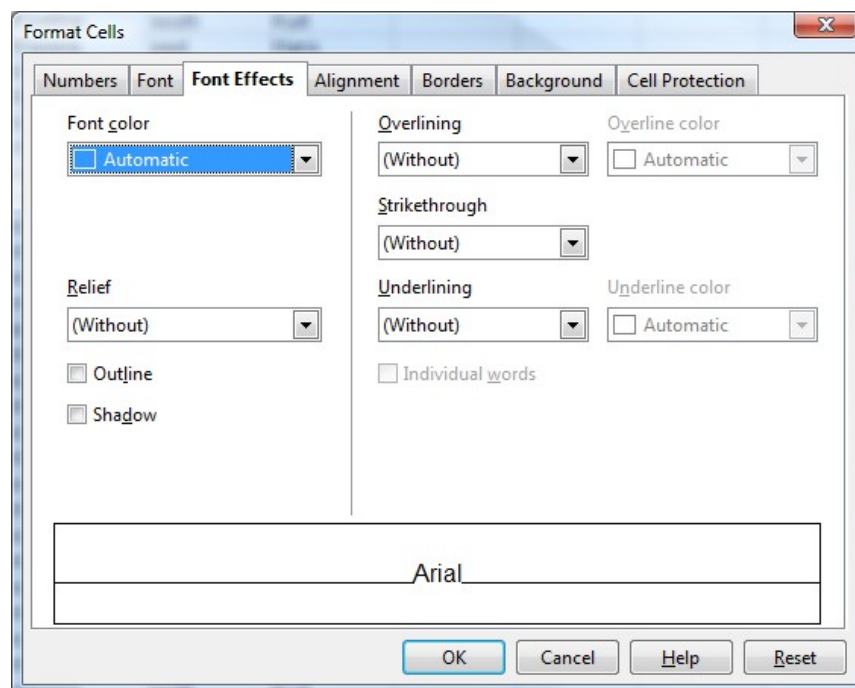


Figure 18: Format Cells > Font Effects

Setting Cell Alignment and Orientation

Some cell alignment and orientation icons are not shown by default on the Formatting toolbar. To show them, click on the small arrow at the right-hand end of the toolbar and select them from the list of icons. Most of the icons are also available on the Sidebar in the alignment panel of the Properties deck.

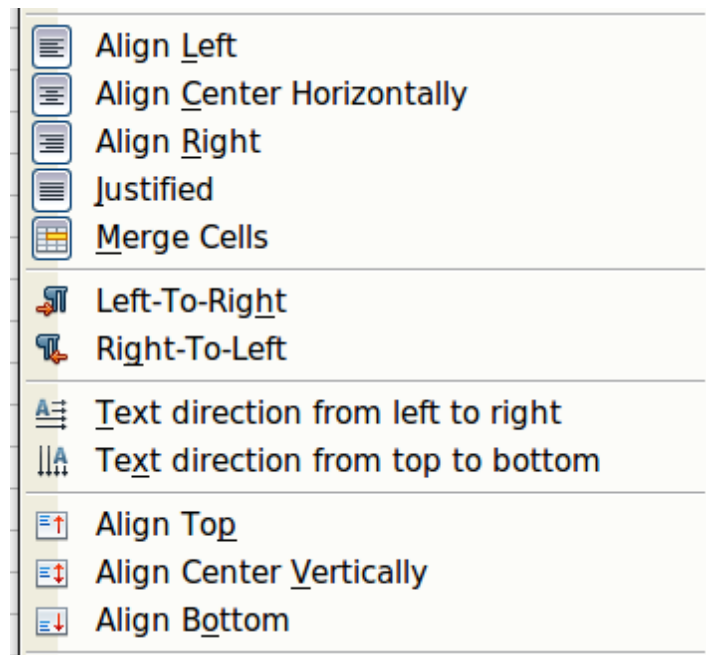


Figure 19: Cell alignment and orientation

Some of the alignment and orientation icons are available only if you have Asian or CTL (Complex Text Layout) languages enabled (in **Tools > Options > Language Settings > Languages**). If you choose an unavailable icon from the list, it will not appear in the toolbar.

For more control and other choices, use the *Alignment* tab (Figure 13) of the Format Cells dialog to set the horizontal and vertical alignment and rotate the text. If you have Asian languages enabled, the *Text orientation* section shows an extra option (labeled **Asian layout mode**) under the **Vertically stacked** option, as shown in Figure 20.

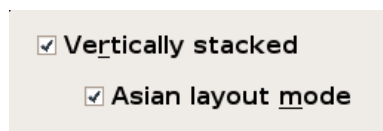


Figure 20: Asian layout mode option

The difference in results between having **Asian layout mode** on or off is shown in Figure 21.

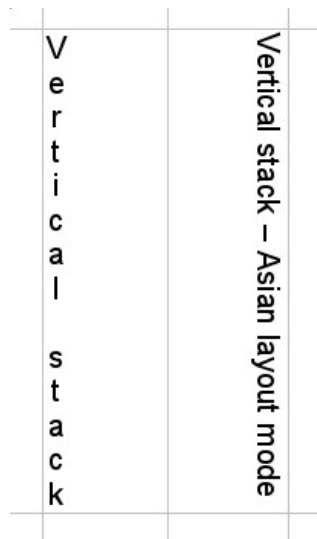


Figure 21: Types of vertical stacking

Formatting the Cell Borders

To choose a line style and color for the borders of a cell quickly, click the small arrows next to the **Line Style** and **Line Color** icons on the Cell Appearance panel of the Sidebar. In each case, a palette of choices is displayed. These are also available on the Formatting toolbar. If the **Line Style** and **Line Color** icons are not shown in the formatting toolbar, select the down arrow on the right side of the bar, then select **Visible Buttons**.

For more control, including the spacing between the cell borders and the text, use the *Borders* tab of the Format Cells dialog. This Dialog also allows you to define a shadow. See Chapter 4 for details.

Note

Cell border properties apply to a cell and can only be changed if you edit that cell. For example, if cell C3 has a top border (which would be equivalent visually to a bottom border on C2), that border can only be removed by selecting C3. It cannot be removed in C2.

Formatting the Cell Background

To quickly choose a background color for a cell, click the small arrow next to the **Background Color** icon on the Sidebar or the Formatting toolbar. A palette of color choices, similar to the Font Color palette, is displayed.

(To define custom colors, use **Tools > Options > OpenOffice > Colors**. See Chapter 14 for more information.)

You can also use the *Background* tab of the Format Cells dialog. See Chapter 4 for details.

Autoformatting Cells and Sheets

Use the AutoFormat feature to quickly and easily apply a set of cell formats to a sheet or a selected cell range.

- 1) Select the cells you want to format, including the column and row headers.
- 2) Choose **Format > AutoFormat**.
- 3) To select which properties (number format, font, alignment, borders, pattern, autofit width, and height) to include in an AutoFormat, click **More**. Select or deselect the required options. Click **OK**.

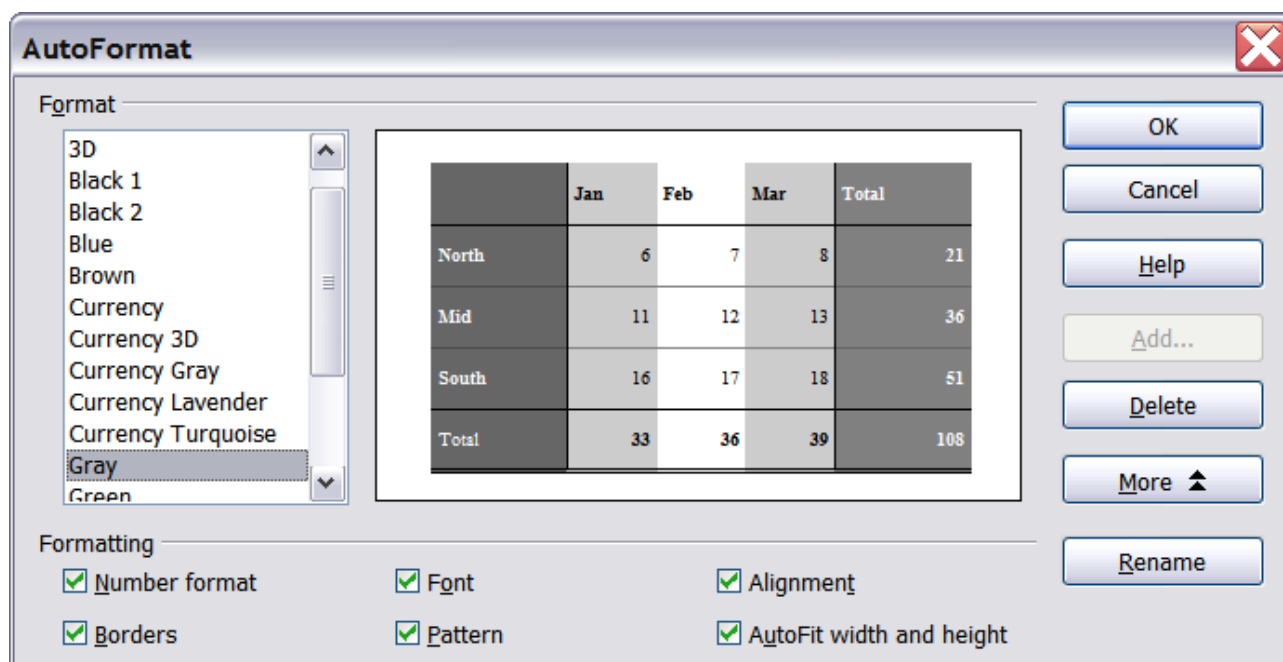


Figure 22: Choosing an AutoFormat

If you do not see any change in the color of the cell contents, choose **View > Value Highlighting** from the menu bar.

Note

If the selected cell range does not have column and row headers, AutoFormat is not available.

Defining a New AutoFormat

You can define a new AutoFormat that's available to all spreadsheets.

- 1) Format a sheet.
- 2) Choose **Edit > Select All**.
- 3) Choose **Format > AutoFormat**. The **Add** button is now active.
- 4) Click **Add**.
- 5) In the *Name* box of the Add AutoFormat dialog, type a meaningful name for the new format.
- 6) Click **OK** to save. The new format is now available in the *Format* list in the AutoFormat dialog.

Formatting Spreadsheets Using Themes

Calc comes with a predefined set of formatting themes you can apply to your spreadsheets.

It is not possible to add themes to Calc, and they cannot be modified. However, you can modify the styles of themes after you apply them to a spreadsheet.

To apply a theme to a spreadsheet:

- 1) Click the **Choose Themes** icon in the Tools toolbar. If this toolbar is not visible, you can show it using **View > Toolbars > Tools**. The Theme Selection dialog appears. This dialog lists the available themes for the whole spreadsheet.



- 2) In the Theme Selection dialog, select the theme you want to apply. As soon as you select a theme, some of the properties of the custom styles are applied to the open spreadsheet and are immediately visible.
- 3) Click **OK**. You can now go to the Styles and Formatting window to modify specific styles. These modifications do not change the theme; they only change the appearance of this specific spreadsheet document.

Using Conditional Formatting

You can set up cell formats to change based on conditions that you specify. For example, in a table of numbers, you can show all the values above the average in green and all those below the average in red.

Note	To apply conditional formatting, AutoCalculate must be enabled. Choose Tools > Cell Contents > AutoCalculate .
-------------	---

Conditional formatting depends on the use of styles. If you are not familiar with styles, please refer to Chapter 4. An easy way to set up required styles is to format a cell the way you want it and click the **New Style from Selection** icon in the Styles and Formatting window. You can define new cell styles before setting up conditional formatting or do it from within the conditional formatting dialog.

To set up conditional formatting:

- 1) In your spreadsheet, select the cells to which you want to apply conditional formatting. You may need to know which cell is the active cell to set up the formatting. If you select several cells, note which cell is active afterward.
- 2) Choose **Format > Conditional Formatting** from the menu bar.
- 3) On the Conditional Formatting dialog (Figure 23), enter the conditions. Click **OK** to save. The selected cells are now formatted in the relevant style.

Cell value is / Formula is

Specifies whether conditional formatting is dependent on a *cell value* or *formula*. If you select *cell value is*, the **Cell Value Condition** box is displayed, as shown in the example. With this box, you can choose from conditions including *less than*, *greater than*, *between*, and others.

Parameter field

Enter a cell reference, value, or formula in the parameter field or in both parameter fields if you have selected a condition requiring two parameters. You can also enter formulas containing relative references. The relative references are defined with respect to the active cell if multiple cells are selected. For example, if you selected the range A1:A10 by dragging down from A1, then A10 will be the active cell. Any relative references should be written as would be appropriate for A10. The reference will be adjusted automatically for the other cells in the range.

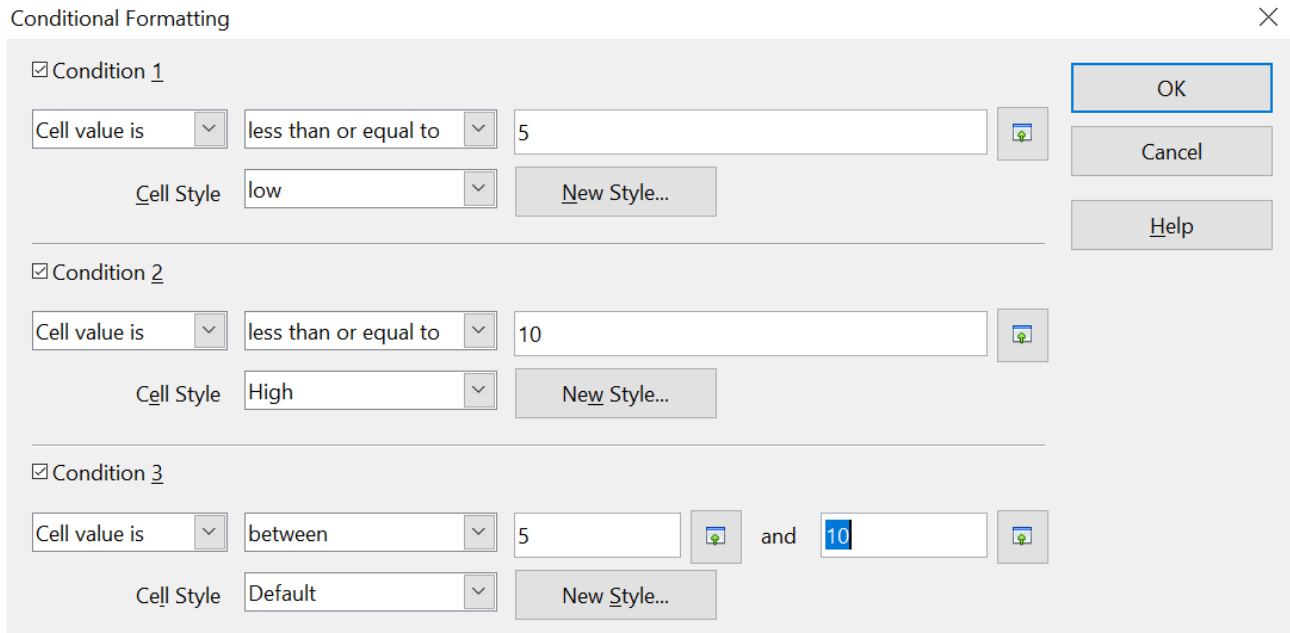


Figure 23: Conditional formatting dialog

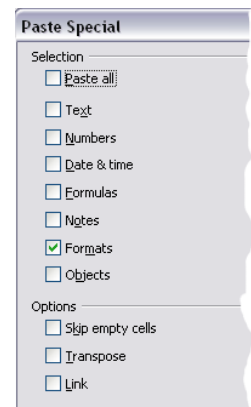
Cell style

Choose the cell style to be applied if the specified condition matches. You can define the style by clicking on the *New Style* button.

See the Help for more information and examples of use.

To apply the same conditional formatting later to other cells:

- 1) Select one of the cells that have been assigned conditional formatting.
- 2) Copy the cell to the clipboard.
- 3) Select the cells that are to receive this same formatting.
- 4) Choose **Edit > Paste Special**.
- 5) On the Paste Special dialog, in the Selection area, select *only* the **Formats** option. Make sure all other options are *not* selected. Click **OK**.



Hiding and Showing Data

When elements are hidden, they are neither visible nor printed but can still be selected for copying if you select the elements around them. For example, if column B is hidden, it is copied when you select columns A and C. When you need a hidden element again, you can reverse the process and show the element.

To hide or show sheets, rows, and columns, use the options on the Format menu or the right-click (context) menu. For example, to hide a row, select the row and then choose **Format > Row > Hide** (or right-click and choose **Hide**).

To hide or show selected cells, choose **Format > Cells** from the menu bar (or right-click and choose **Format Cells**). On the Format Cells dialog, go to the *Cell Protection* tab.

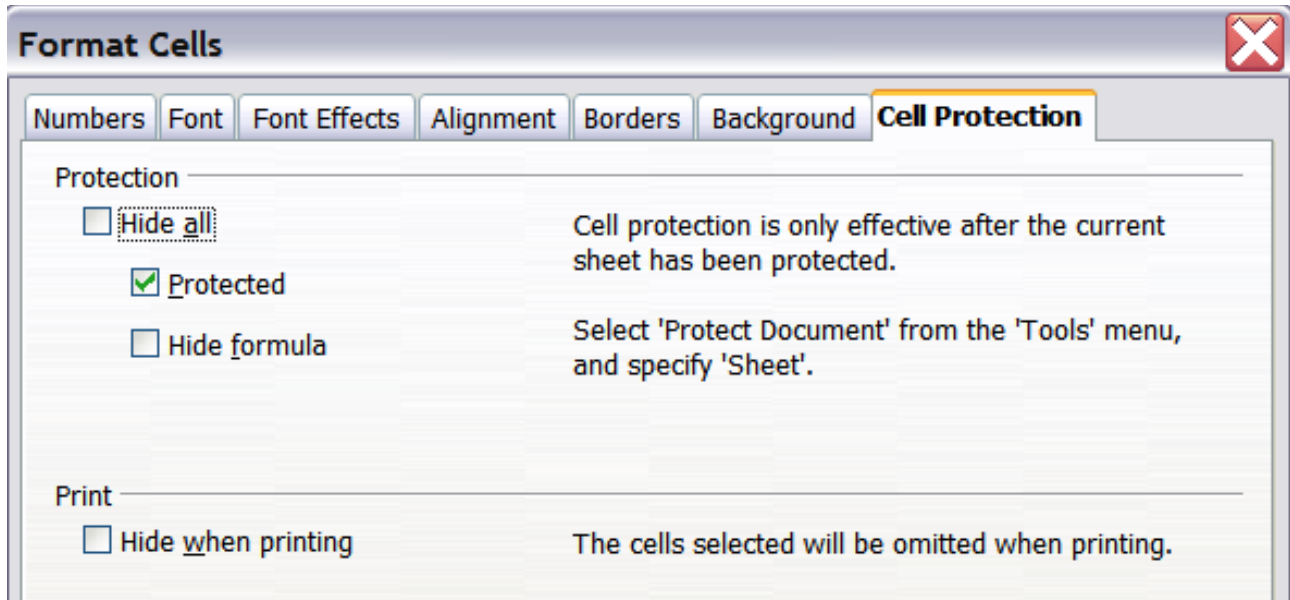


Figure 24: Hiding or showing cells

Outline Group Controls

If you continually hide and show the same cells, you can simplify the process by creating *outline groups*. These groups add a set of quick-to-use and always-available controls for hiding and showing the cells in the group.

If the contents of cells fall into a regular pattern, such as four cells followed by a total, then you can use **Data > Group and Outline > AutoOutline** to have Calc add outline controls based on the pattern. Otherwise, you can set outline groups manually by selecting the cells for grouping, then choosing **Data > Group and Outline > Group**. On the Group dialog, you can choose whether to group the selected cells by rows or columns.

When you close the dialog, the outline group controls are visible between the row or column headers and the edges of the editing window. In appearance, the controls resemble the tree structure of a file manager. They can be hidden by selecting **Data > Group and Outline > Hide Details**. They are strictly for interactive use and do not print.

The basic outline controls have plus or minus signs at the start of the group to show or hide hidden cells. However, if outline groups are nested, the controls have numbered buttons for hiding the different levels.

If you no longer need a group, place the mouse cursor in any cell and select **Data > Group and Outline > Ungroup**. To remove all groups on a sheet, select **Data > Group and Outline > Remove**.

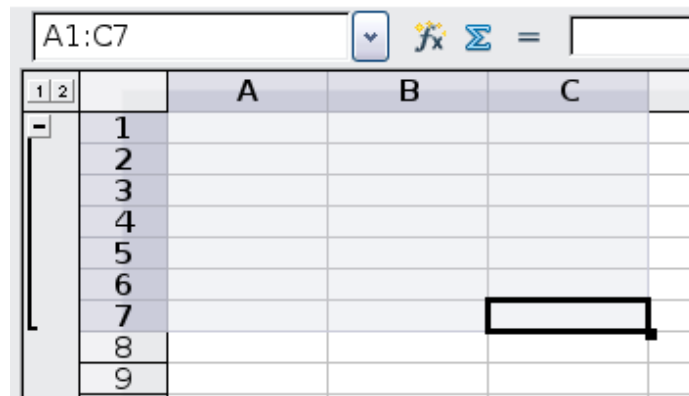


Figure 25: Outline group controls

Filtering Which Cells are Visible

A filter is a list of conditions that each entry must meet to be displayed. You can set three types of filters from the **Data > Filter** sub-menu.

Automatic filters add a drop-down list to the top row of a selected range, often the whole column, that contains commonly used filters. They are quick, convenient, and almost as useful with text as with numbers because the list includes every unique entry in the selected cells.

In addition to these unique entries, automatic filters include the option to display all entries, the ten highest numerical values, and all cells that are empty or not empty, as well as a standard filter that you can customize (see below). However, they are somewhat limited. In particular, they do not allow regular expressions, so you cannot use them to display cell contents that are similar but not identical.

Standard filters are more complex than automatic filters. You can set as many as three conditions as a filter, combining them with the operators *AND* and *OR*. Standard filters are primarily useful for numbers, although a few of the conditional operators, such as = (equal) and <> (not equal) can also be used for text. Regular expressions can be turned on under *More Options* so that text can be matched very flexibly.

Other conditional operators for standard filters include options to display the largest or smallest values or a percentage of them. Although useful in themselves, standard filters take on added value when they are used to further refine automatic filters.

The results of the Standard filter can be copied to another location, rather than displayed in place, by hiding rows. This is very useful for creating subsets of data.

Advanced filters are structured similarly to standard filters. The differences are that advanced filters are not limited to three conditions, and their criteria are not entered in a dialog. Instead, advanced filters are entered in a blank area of a sheet and then referenced by the advanced filter tool to apply them.

Figure 26 shows a simple data set in columns A through D and filter criteria for it in columns F through I. Filter conditions that appear in the same row are connected by *AND*, and conditions in different rows are connected by *OR*. The conditions displayed will get data for (*Bob & February & Price >6*) *OR* (*Ann & Pear & Price <= 5*). The filter is activated with **Data > Filter > Advanced Filter**, where you can set the cell range containing the filter conditions. The cell range should include the first row with the headers, e.g., F1:I3. Remember that the filter conditions in the figure are positioned so that applying the filter will hide some of the rows displaying the

conditions. That does not impair the filter but might prove inconvenient if you want to read the filter conditions.

A	B	C	D	E	F	G	H	I
Client Name	Month	Product	Price		Client Name	Month	Product	Price
Ann	January	Apple	5		Bob	February		> 6
Bob	February	Apple	10		Ann		Pear	<= 5
Ann	March	Apple	15					
Bob	January	Apple	5					
Ann	February	Apple	10					
Ann	March	Apple	15					
Bob	January	Apple	5					
Bob	February	Pear	10					
Bob	March	Pear	15					
Ann	January	Pear	5					
Bob	February	Pear	10					
Ann	March	Pear	15					

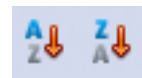
Figure 26: Advanced filter conditions for a data set before applying the filter

Sorting Records

Sorting rearranges the visible cells on the sheet. In Calc, you can sort by up to three criteria applied one after another. Sorts are handy when searching for a particular item and become even more powerful after filtering data.

In addition, sorting can be useful when adding new information. When a list is long, adding new information at the bottom of the sheet is usually easier than inserting rows in the proper places. After you have added the information, you can sort it to update the sheet.

Highlight the cells to be sorted, then select **Data > Sort** to open the Sort dialog (Figure 27) or click the **Sort Ascending** or **Sort Descending** toolbar buttons. Using the dialog, you can sort the selected cells using up to three columns in ascending (A-Z, 1-9) or descending (Z-A, 9-1) order.



Tip

You can define a custom sort order if the supplied alphanumeric ones do not fit your requirements. See “Defining a Fill Series” on page 9 for instructions.

Sort

Sort Criteria
Options

Sort by

Client Name

☒ Ascending
☐ Descending

Then by

Month

☒ Ascending
☐ Descending

Figure 27: Choosing the criteria and order of sorting

On the *Options* tab of the Sort dialog (Figure 28), you can choose any of the following:

Figure 28: Options for sorting

Case sensitive

If two entries are otherwise identical, one with an upper case letter is placed before one with a lower case letter in the same position if the sort is descending. If the sort is ascending, the entry with an upper case letter is placed after one with a lower case letter in the same position.

Range contains column labels

Does not include the column heading in the sort.

Include formats

A cell's formatting is moved with its contents. If formatting is used to distinguish different types of cells, then use this option.

Copy sort results to

Sets a spreadsheet address to which to copy the sort results. If a specified range does not have the necessary number of cells, cells are added. The sort fails if a range contains cells that already have content.

Custom sort order

Select the box, then choose one of the sort orders defined in **Tools > Options > OpenOffice Calc > Sort Lists** from the drop-down list.

Direction

Sets whether rows or columns are sorted. The default is to sort rows.

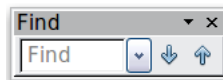
Finding and Replacing in Calc

Calc has two ways to find text within a document:

- the Find toolbar for fast text searching
- the Find & Replace dialog

Using the Find Toolbar

The Find toolbar is located by default on the right-hand end of the Standard toolbar. You can hide or show the Find toolbar using **View > Toolbars > Find**.



Type a search term in the Find box, and then click the Find Next (down-arrow) or Find Previous (up-arrow) button. To find other occurrences of the same term, continue clicking the button.

Using the Find & Replace Dialog

To display the **Find & Replace** dialog (Figure 29), use the keyboard shortcut **Control+F** or choose **Edit > Find & Replace** from the menu bar.

In spreadsheet documents, you can search for text, formulas, and styles. You can navigate from one occurrence to the next using **Find**, or you can highlight all matching cells at once using **Find All**, then apply another format or replace the cell contents with other content.

Text and numbers in cells may have been entered directly or may be the result of a calculation. The search method you use depends on the data type you're searching for.

Tip

Cell contents can be formatted in different ways. For example, a number can be formatted as currency to be displayed with a currency symbol such as the dollar sign (\$). You will see the currency symbol in the cell, but you can't search for it.

By default, Calc searches the current sheet. To search through all document sheets, click **More Options**, then select *Search in all sheets* option.

Caution



To avoid the possibility of mistakes, use **Replace All** with caution. If it's not discovered promptly, a mistake with **Replace All** might require a laborious, time-consuming, manual, word-by-word search to fix.

Find & Replace ✕

Search for

Find
Find All

Replace with

Replace
Replace All

☐ Match case
☐ Entire cells

Fewer Options ▲

Help

Close

☐ Current selection only
☐ Backwards
☐ Regular expressions
☐ Similarity search
☐ Search for Styles

Search in: Formulas

Search direction:
☒ Rows
☐ Columns

☐ Search in all sheets

Figure 29: Expanded Find & Replace dialog

Finding and Replacing Formulas or Values

You can use the Find & Replace dialog to search in formulas or the displayed values resulting from a calculation.

- 1) To open the Find & Replace dialog, use the keyboard shortcut *Control+F* or select **Edit > Find & Replace**.
- 2) Click **More Options** to expand the dialog.
- 3) Select *Formulas* or *Values* in the *Search in* drop-down list.
 - Formulas finds parts of the formulas.
 - Values finds the results of the calculations.
- 4) Type the text you want to find in the *Search for* box.

- 5) To replace the text with different text, type the new text in the *Replace with* box.
- 6) When you have set up your search, click **Find**. To replace text, click **Replace** instead.

Finding and Replacing Text

- 1) Open the Find & Replace dialog, click **More Options** to expand the dialog, and select *Values* or *Notes* in the *Search in* drop-down list.
- 2) Type the text you want to find in the *Search for* box.
- 3) To replace the text with different text, type the new text in the *Replace with* box.
- 4) Click **Find**, **Find All**, **Replace**, or **Replace All**.

When you click **Find**, Calc selects the next cell that contains your text. You can edit the text, then click **Find** again to advance to the next found cell. If you closed the dialog, you can press *Ctrl+Shift+F* to find the next cell without opening the dialog.

When you click **Find All**, Calc selects all cells that contain your entry. Now, you can apply a cell style to all of them simultaneously.

Finding and Replacing Cell Styles

To quickly change all the paragraphs of an unwanted style to another preferred style:

- 1) On the expanded Find & Replace dialog, select **Search for Styles**. The *Search for* and *Replace with* boxes now contain a list of styles.
- 2) Select the styles you want to search for and replace.
- 3) Click **Find**, **Find All**, **Replace**, or **Replace All**.

Using Wildcards (Regular Expressions)

Wildcards, one form of what's known as *regular expressions*, are a character or combination of characters that describe to AOO a pattern of text. For example, the regular expression *\d* represents any of the numeric characters 0 - 9. The full set of regular expressions is too large to explain here, but we'll present a few tips about using them in Calc. The table below describes the regular expressions used in the following examples.

Regular Expression	Meaning
.	Any character
+	One or more of the preceding character. So, .+ means one or more of any character.
?	Zero or one of the preceding character. The expression an? would match both a and an because the ? allows for zero or one n after the a.
\n	A hard line break within a cell.
\$	The end of a paragraph of text. The expression xls\$ matches xls if is at the end of the paragraph, as it would be in a list of filenames with one filename per cell.
^	The beginning of a paragraph.
&	In the <i>Replace with</i> box of the Find & Replace dialog, this represents the text found by the search.

To use wildcards and regular expressions when searching and replacing:

- 1) On the Find & Replace dialog, click **More Options** to see more choices. On this expanded dialog, select the **Regular expressions** option.
- 2) Type the search text, including the wildcards, in the *Search for* box and the replacement text (if any) in the *Replace with* box.
- 3) Click **Find**, **Find All**, **Replace**, or **Replace All** (not recommended).

Tip The online help describes many of the regular expressions and their uses.

The following points may be of interest to Calc users:

- In Calc, regular expressions are applied separately to each cell. This means that a search for **r.d** will match **red** in cell A1 but not **r** in cell A2 and **d** (or **ed**) in cell A3. (The regular expression **r.d** means the system will try to match **r** followed by any other character followed by **d**.)
- When a match is found, the entire cell is highlighted, but only the text found will be replaced. For example, searching for **brown** will highlight a cell containing **redbrown clay**, and choosing nothing in the *Replace with* box leaves the cell containing **red clay**.
- If Find is used twice in a row, and the second time the **Current selection only** box is activated, then the second search will evaluate the *whole* of each selected cell, *not* just the strings that caused the cells to be selected in the first search. For example, searching for **joh?n**, then activating **Current selection**

only and searching for **sm.th** will find cells containing **Jon Smith** and **Smythers, Johnathon**.

- If a cell contains a hard line break (entered by *Ctrl+Enter*), it may be found by using the regular expression **\n**. For example, if a cell contains **red[hard line break]clay**, searching for **d\n**c and entering nothing in the *Replace with* box leaves the cell containing **relay**.
- The hard line break acts to mark “end of text” (similar to “end of paragraph” in Writer), found by the regular expression special character **\$**, in addition to the end of text in the cell. For example, if a cell contains **red [hard line break] clay**, then a search for **d\$** replacing it with **al** leaves the cell with **real [hard line break] clay**. Note that with this syntax, the hard line break is not replaced.
- Using **\n** in the *Replace with* box will replace with the literal characters **\n**, not a hard line break.
- The Find & Replace dialog has an option to search within formulas, values, or notes. This option applies to any search, not just one using regular expressions. Searching with the *Formulas* option for **SUM** would find a cell containing the formula **=SUM(A1:A6)** and a cell containing the simple text **SUMMARY**. If you search in the *Values* for **SUM**, you will find the text **SUMMARY** but not the cell with the **SUM()** formula because the cell value will be a number.
- Searching for the regular expression **^\$** will not find empty cells. This is intentional to avoid performance issues when selecting a huge number of cells. Note that empty cells will not be found even if you only search a selection.
- Finding cell contents using the regular expression **.+** (or similar) and replacing them with **&** effectively re-enters the cell contents without any formatting. This technique can remove Calc's automatically applied formatting when importing data from the clipboard or badly formatted files. For example, to convert text strings consisting of digits into actual numbers, format the cells as numbers, and then perform the search and replace.

See Chapter 7 (Using Formulas and Functions) for the use of regular expressions within formulas. For now, just know that a regular expression is a sequence of characters that can be used as shorthand for a larger sequence or set of sequences.